

Problem 5.13

```
%{  
AERE 1610: Nathaniel Moberg, Lab4 Problem 5.13  
purpose: write a function that will receive a matrix as a function and  
return the average of all numbers in it  
%}  
A = input('Input a matrix e.g. [x,y,z]: '); %sets output to an input matrix  
average = MatrixAverage(A); %set average equal to function applied to matrix A  
fprintf('the average of the matrix you entered is %g', average) %disp average
```

```
function [average] = MatrixAverage(A)  
%MATRIXAVERAGE takes in matrix as an input and outputs the average of all elements added up  
arguments (Input)  
A  
end  
arguments (Output)  
average  
end  
[m,n] = size(A); %pulls values for dimensions of A  
sum = 0; %initializes sum to 0  
for i = 1:m %for every row...  
for j = 1:n %for every column in every row...  
sum = sum + A(i, j); %add the number at the matrix coordinate to all previous numbers  
end %end block  
end %end block  
average = sum/(m*n); %average is the total sum of all numbers divided by the size of A (the number of rows *  
number of columns)  
end
```

Output:

Problem 6.2

```
% Write a function that receives an input argument x and returns y and z, which should be calculated as follows:  
%  $y = 2x^2$   $z = 3x - 4y$   
function [y, z] = myFirstProgram(x)  
  
y = 2*(x^2);  
z = (3*x)-(4*y);  
  
end
```

Output :

```
[y, z] = MyFirstProgram(5)
```

```
y =
```

```
50
```

```
z =
```

```
-185
```

Problem 6.5

Given the following function header:

```
function doit(a, b)
```

Which of the following function calls would be valid – and why?

a) `fprintf('The result is %.1f\n', doit(4,11))`

b) `doit(5, 2, 11.11)`

c)

```
x = 11;
```

```
y = 3.3;
```

```
doit(x,y)
```

Answer C is correct because it takes two inputs but doesn't expect an output as specified in the function header. Answer A expects an output that it can round to two decimal places which is incorrect, and answer B tries to put in 3 inputs which is incorrect.

Problem 6.14

```
clear, clc
```

```
%{
```

```
Frank Sucansky, Nathan Moberg, Luis Duran, AERE 1610 Lab 4 Problem 6.14
```

```
Write a script that will:
```

- Call a function to prompt the user for an angle in degrees
- Call a function to calculate and return the angle in radians. (Note: π radians = 180°)
- Call a function to print the result

```
Write all of the functions, also. Put the script and all functions in separate code files.
```

```
%}
```

```
deg = user_input(); %user input takes no input as a function, it runs when the file is executed and returns  
%the users input of degrees
```

```
rad = deg2rad(deg); %takes in user input degrees and converts to radians
```

```
print(deg, rad); %takes in calculated radians and displays them to the screen
```

```
function [degrees] = user_input()
```

```
%user input takes no input and returns the users input of degrees
```

```
arguments (Output)
```

```
degrees
```

```
end
```

```
degrees = input('enter an angle in degrees: ');
```

```
end
```

```
function [radians] = deg2rad(degrees)
```

```
%DEG2RAD takes in an angle in degrees and returns that angle in radians
```

```
arguments (Input)
```

```
degrees
```

```
end
```

```
arguments (Output)
```

```
radians
```

```
end
```

```
radians = pi * degrees / 180; %180 degrees = pi radians
```

```
end
```

```
function print(degrees, radians)
```

```
%PRINT Summary of this function goes here
```

```
% Detailed explanation goes here
```

```
arguments (Input)
```

```
degrees
radians
end
fprintf('%g degrees converted to radians is %g radians \n', degrees, radians)
%print message displaying degrees converted to radians
end
```

Output :

```
enter an angle in degrees: 180
180 degrees converted to radians is 3.14159 radians
>>
```